Xi'an Jiaotong-Liverpool University
西交利物浦大學

Module Code: CHE105
Module Title: Introductory Organic Chemistry for Non-Chemists
Semester 1, 2021/22 Year 2

FINAL EXAMINATIONS

Duration: 2 hours

Module Examiner: Dr. E.Amigues **Department: Chem. Tel: 0512 8816 7123**

Instructions to Candidates:

1. Answer all questions.

2. The maximum mark is 100.

3. You can answer the questions in any order you prefer, e.g. start from Q4.

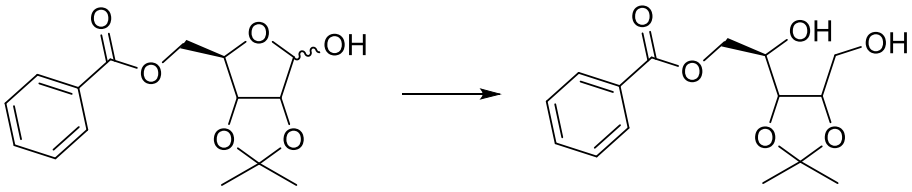
4. Answers should be written on the Answer Booklet provided; the left-hand page of the Answer Booklet can be used for drafting work when necessary.

5. Answers should be written using pen, black or blue, **NO PENCIL**.

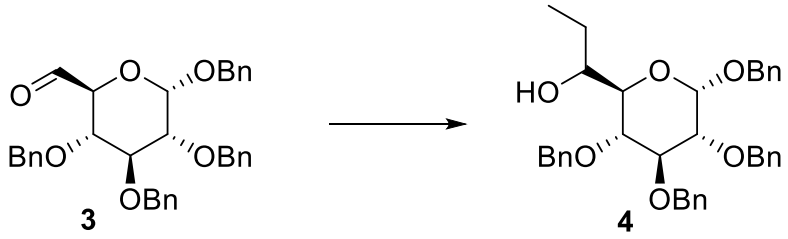
6. Only solutions written in English are accepted.

Question 1 (10 marks)

Identify the suitable reagents you should use in order to perform the following transformations and draw the mechanisms. (2 marks for the reagents, 3 marks for the mechanism)



1 2

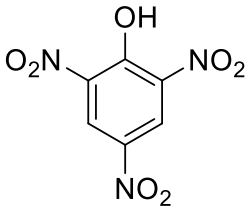


3 4

Question 2 (8 marks)

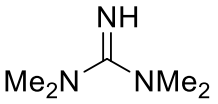
Give explanations for the following observations (2 marks per explanation)

a. In aqueous solution picric acid is a much stronger acid (pKa ≈ 0.3) than phenol (pKa ≈ 10).

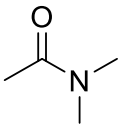


Picric acid

b. Tetramethyl guanidine (pKa≈ 12) is a much stronger base than *N*,*N*-dimethylacetamide (pKa≈ -0.5).

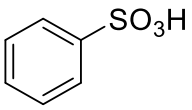


tetramethyl guanidine

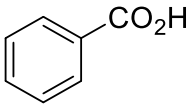


N,N-dimethylacetamide

c. It has been found that benzenesulfonic acid (pKa≈ -6) is a much stronger acid than benzoic acid (pKa≈ 5). Discuss the factors that can account for this.

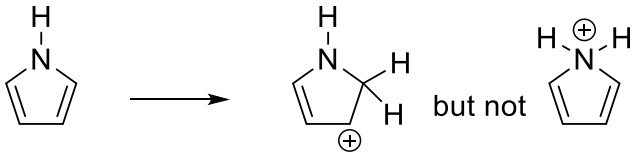


Benzenesulfonic acid



Benzoic acid

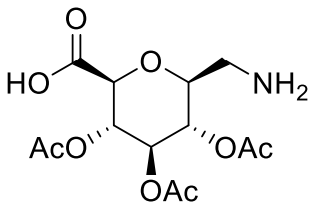
d. The reaction of pyrrole with a strong acid leads to protonation on C-2 not on the nitrogen. Explain why.



Question 3 (20 marks)

Identify the products **A** and **B** that will be formed in reactions below and draw the mechanisms. (10 marks for each reaction, 4 marks for the product, 6 marks for the mechanism)

Reaction 1

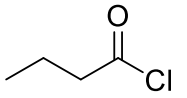


$Ac_2O/AcOH$

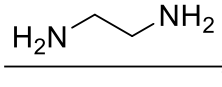
$C_{15}H_{21}NO_{10}$

A

Reaction 2



2 equivalents

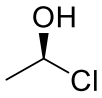


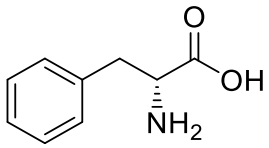
$C_{10}H_{20}N_2O_2$

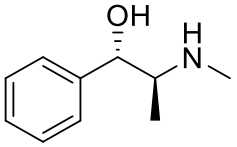
B

Question 4 (12 marks)

Assign the configuration (*R* or *S*) to each of the stereogenic centres in the three molecules below. Clearly show how you have worked out your answer (6 marks for the assignment, 6 marks for the justification)



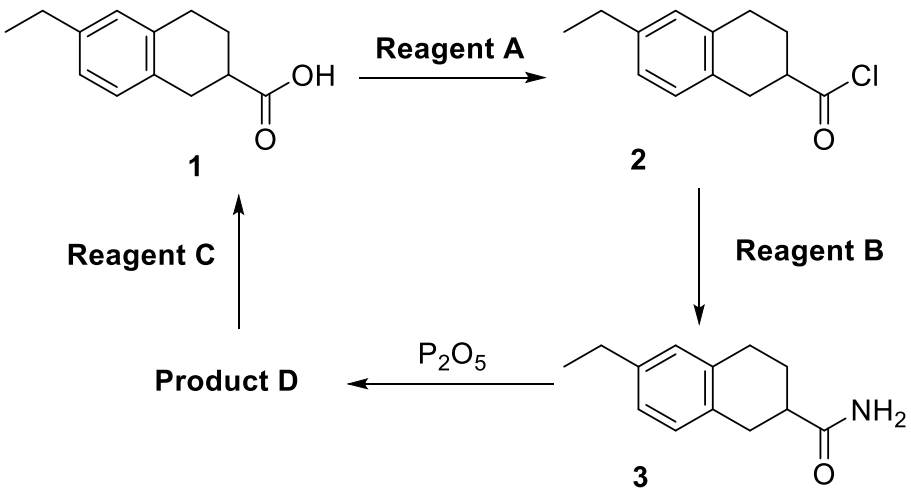




Show the CIP priority rules for every single stereocenter

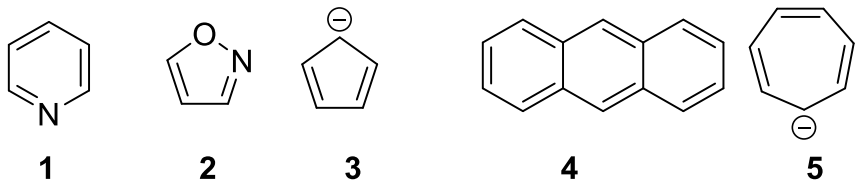
Question 5 (10 marks)

Identify the reagents **A, B, C** and the product **D**. No mechanisms required. (2.5 marks each)



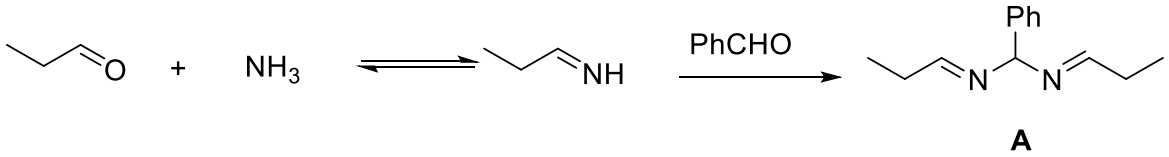
Question 6 (10 marks)

Which structural features does a compound need to have in order to be aromatic? (5 marks). Which of these compounds are aromatic? Justify your answer with electrons counting. (5 structures, 1 marks each)



Question 7 (20 marks)

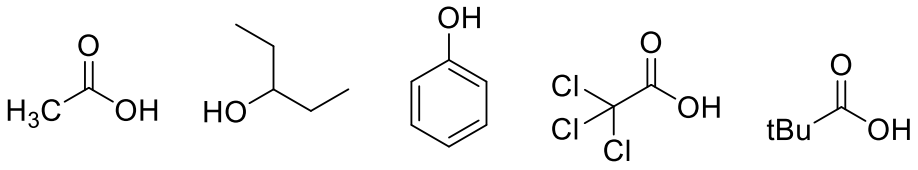
Suggest mechanisms for these reactions that lead to the formation of product **A**. (20 marks)



A

Question 8 (10 marks)

Rank compounds **A** to **E** in order of increasing acidity. Explain your answers.



A **B** **C** **D** **E**

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